

Computational Work 2: Cholesky Factorization

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Course: Computational Matrix Methods

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Abstract

This work investigates two computational problems involving numerical linear algebra and the numerical solution of a one-dimensional steady-state heat conduction problem. First, a symmetric matrix generated according to the prescribed procedure is analyzed to determine whether it is positive definite, followed by the computation of its Cholesky factorization, condition number, and the solution of the associated linear system. Second, the steady-state heat conduction equation in a unit-length bar is discretized using the finite difference formulation specified in the problem statement, resulting in a linear system whose solution provides the temperature distribution along the bar. The numerical results are analyzed with respect to the algebraic properties of the matrices, the conditioning of the linear systems, and the physical interpretation of the computed temperature profile. All problem-specific data are generated deterministically from the student's enrollment number according to the procedures provided in the assignment.